



Problem-Based Learning in Medical Education: A Global Research Landscape of the Last Ten Years (2013–2022)

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Abstract

Background Problem-based learning (PBL) constructs a curriculum that merges theory and practice by employing clinical scenarios or real-world problems. Originally designed for the pre-clinical phase of undergraduate medicine, PBL has since been integrated into diverse aspects of medical education. Therefore, this study aims to map the global scientific landscape related to PBL in medical education in the last ten years.

Methods We combined bibliometrics and network analysis to analyze the metadata of related research articles published between 2013 and 2022 and indexed in the Web of Science Core Collection.

Results Our results show an annual publication rate of 9.42%. The two main journals disseminating research on this subject are *BMC Medical Education* and *Medical Teacher*. Education & Educational Research and Health Care Sciences & Services are the two most frequent research areas, and also the two most central nodes of the related network. The USA and China are the most publishing countries, while the Netherlands and Canada are the most collaborative. The Maastricht University holds the position of most publishing and collaborative research organization. The University of California ranks second in publications, while the University of Toronto is the second most central research organization.

Conclusions Our study provides an overview of the last ten years of publications related to PBL and medical education, and we hope it can be of interest to educators, researchers, and students involved with this subject.

Keywords Problem-based learning · Medical education · Scientific publication · Bibliometrics · Network analysis

Introduction

First introduced at McMaster University in Canada in the late 1960s, problem-based learning (PBL) soon garnered worldwide attention for its unique approach to medical education [1, 2]. In the 1980s, it was adopted by Maastricht University in the Netherlands, which quickly became a global leader in employing this methodology [3]. Decades later, it is now adopted in more than 500 universities worldwide [4], including in Asia [5–7], Africa [8], and the Middle East [9].

Unlike traditional medical education models, PBL builds a curriculum that promotes a learning environment that integrates practice and theory through clinical scenarios or real-world problems [10–12]. Students actively investigate and solve complex medical problems, reinforcing their theoretical knowledge through practical application [13–15].

Nowadays, it can be considered that PBL has at least four essential characteristics: (a) the use of engaging tasks or

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authentic problems as a starting point for learning; (b) self-directed and self-regulated learning; (c) working in groups of students tackling these problems in a collaborative and solidary way; (d) and the role of the teachers as a facilitator of this process [12, 16, 17]. Thus, the PBL approach promotes an active learning experience that challenges students to develop essential problem-solving skills while acquiring relevant and immediately applicable knowledge to their future medical practice [13, 18]. This dynamic educational model provides a more immersive, practical, and contextualized learning experience that aligns with the medical profession's evolving demands by preparing students to tackle real-world problems efficiently and effectively [11, 14, 15].

Initially developed for use in the pre-clinical phase of the undergraduate medical curriculum, PBL has since been integrated into various aspects of medical education, showcasing its versatility and adaptability [2, 19]. Recent systematic reviews show that PBL positively affects the academic achievement of undergraduate medical courses, including improving social and communication skills, problem-solving, and self-learning skills [13, 20]. After decades of applying PBL across diverse medical education settings, a broad and diverse scientific literature examining the PBL methodology in the context of medical education is available [4, 19]. The broadness nature of the literature, with its varying perspectives and findings, makes it challenging to gain a comprehensive understanding of the evolution of research in this area.

Thus, what is the scientific landscape related to PBL in medical education? This study aims to map the global scientific landscape related to PBL in medical education in the last ten years. To do so, we combined bibliometrics and network analysis to analyze the metadata of research articles related to this subject published between 2013 and 2022 and indexed in the Web of Science Core Collection (WoS).

Overall, there are many studies somehow examining the PBL methodology in the context of medical education. A search in WoS would show over 400 such articles. There are also two dozen articles that used bibliometrics and/or network analysis to analyze a variety of PBL-related subjects. Examples are the use of these methodologies to identify top-cited articles in PBL and assess their findings [25], investigate online PBL [26], assess the impact of PBL on students [27], and evaluate the use of PBL in chemical engineering [28]. However, none of them focused on medical education (search strategies applied in WoS are available in the Supplementary Information). Thus, our study contributes to the existing knowledge by providing a global landscape of the research produced on the use of PBL in medical education, encompassing the last ten years. In doing so, we hope our results can interest those interested or involved with using PBL in medical education, especially educators, researchers, and students.

Methods

Bibliometrics and network analysis were used to analyze the metadata of research articles related to PBL and medical education. Bibliometrics is a quantitative method that uses statistics to analyze the frequencies and co-occurrences of scientific publication metadata [21], enabling the mapping of trends in science development and collaboration [22]. For its part, network analysis uses mathematics and graph theory to explore and visually represent co-occurrences, patterns, and interactions among the variables within the metadata of scientific publications, encompassing articles and patents [23]. Overall, bibliometrics and network analysis are often used together to provide a more comprehensive understanding of the subject under investigation [24]. The research articles metadata were collected in WoS in April 2023 according to the following search strategy:

TI = ("Problem-Based Learning" or "Problem-Based Curriculum*" or "Experiential Learning" or "Active Learning") and TS = ((medical or medicine or physician*) near/1 (education* or learning or training)) and Article (Document Types)

Timespan: 2013-01-01 to 2022-12-31 (publication date)

The search strategy was applied in the WoS advanced search mode using the tag Title (TI: searches in the documents' titles) for PBL descriptors and the tag Topic (TS: searches in the documents' titles, abstracts, and keywords) for descriptors related to medical education. The PBL descriptors were collected in the Medical Subject Headings (MeSH: <https://www.ncbi.nlm.nih.gov/mesh/>) and combined with the Boolean operator OR, while the medical education descriptors are free-text words combined with the Boolean proximity operator NEAR. We restricted the document types to research articles (Articles) and set the timespan to retrieve articles published in the last ten years.

The search retrieved 231 article records, which were imported into the data/text mining software VantagePoint 11.0 for treatment and analysis. No duplicated records were found. Once the study is limited to human medical education, we then excluded two articles assigned to the research area Veterinary Sciences from the analysis, leaving a total of 229 articles. Research area is a WoS subject classification scheme that indexes all publications in this database (Clarivate: <https://incites.help.clarivate.com/Content/Research-Areas/wos-research-areas.htm>). Next, the authors' affiliations (also referred to as organizations) and authors' keywords were cleaned and standardized using the VantagePoint list cleanup tool (fuzzy matching rule general) along with manual cleaning.

Also, using VantagePoint, we generated co-occurrence matrices of research areas, countries, organizations, and keywords. Then, these matrices were imported into the network analysis software Gephi 0.10, where we built the networks and calculated the network metrics of degree centrality (DC), weighted degree centrality (WDC), and eigenvector centrality (EC). DC is the number of direct connections a node has with other nodes in the network and is an effective measure of prominence and influence, while WDC is the number of node connections weighted by the strength of those. The co-occurrences between the nodes determine the strength of a node's connections. In turn, EC measures the centrality of a node based on its connections to other central nodes in the network. It assigns a higher score to a node if connected to other high-score nodes [23]. Overall, nodes with high centrality values play a strategic role in the network [29]. The Fruchterman-Reingold algorithm gives the network layout (Gephi: <https://github.com/gephi/gephi/wiki/Fruchterman-Reingold>). We used the analysis and graphing software GraphPad Prism 8 to produce the frequency graphs and combinations of networks and graphs.

Limitations of the Study

As known, a search strategy has to balance precision and coverage, and the option for one or another depends on the purpose of the study. In our study, to increase precision and avoid articles whose focus was not PBL, we performed the search using the tag TI for PBL descriptors. In doing so, our study does not include in the analysis all articles indexed in WoS that are somehow related to the use of PBL in medical education, we specifically targeted those directly addressing PBL in the title. Our study also covers only the last ten years of publications. However, this period is coherent with the authors' choice to provide a more current scientific landscape on PBL and medical education. This relatively short timespan is particularly important when performing network

analysis, considering the goal of identifying a more recent picture of institutional collaborations between organizations and co-occurrences between variables such as keywords and research areas. Due to the inclusion of only English-language WoS publications, we recognize the possible linguistic bias in our study. We may have inadvertently overlooked significant contributions and perspectives presented in other languages or in less accessible journals. However, this is common practice in bibliometric studies to maintain the standard and robustness of the analysis. Furthermore, our study predominantly adopts a quantitative approach, potentially overlooking the qualitative dimensions inherent in PBL, such as personal experiences and instructional subtleties. Future studies could explore these qualitative aspects, offering deeper insights into the effectiveness and nuances of PBL in medical education.

Results

Figure 1a shows the annual evolution of scientific publications from January 2013 to December 2022. During this period, the annual growth rate of publications was 9.42%. The number of annual publications peaked in 2019 (38 articles), dropping 34.21% in the next year. However, the last two years show an increase in annual publications, surpassing 30 articles per year in 2022. Figure 1b shows the main journals that have published articles on the topic. The top three are *BMC Medical Education* (8.30% of total publications), *Medical Teacher* (7.42%), and *Medical Science Educator* (4.80%).

Figure 2a depicts the most frequent research areas. Education & Educational Research was the most frequent (54.58% of total publications), followed by Health Care Sciences & Services (19.65%), and General & Internal Medicine (12.66%). The first research area is also the most central node in the network of research areas (Fig. 2b) according

Fig. 1 Publication years and journals (2013–2022). **a** Evolution of annual publications. **b** Journals with four or more articles

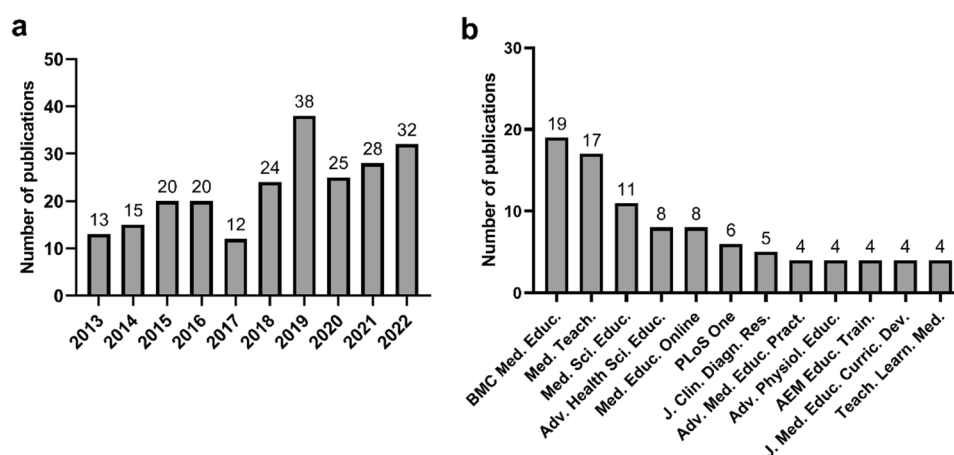
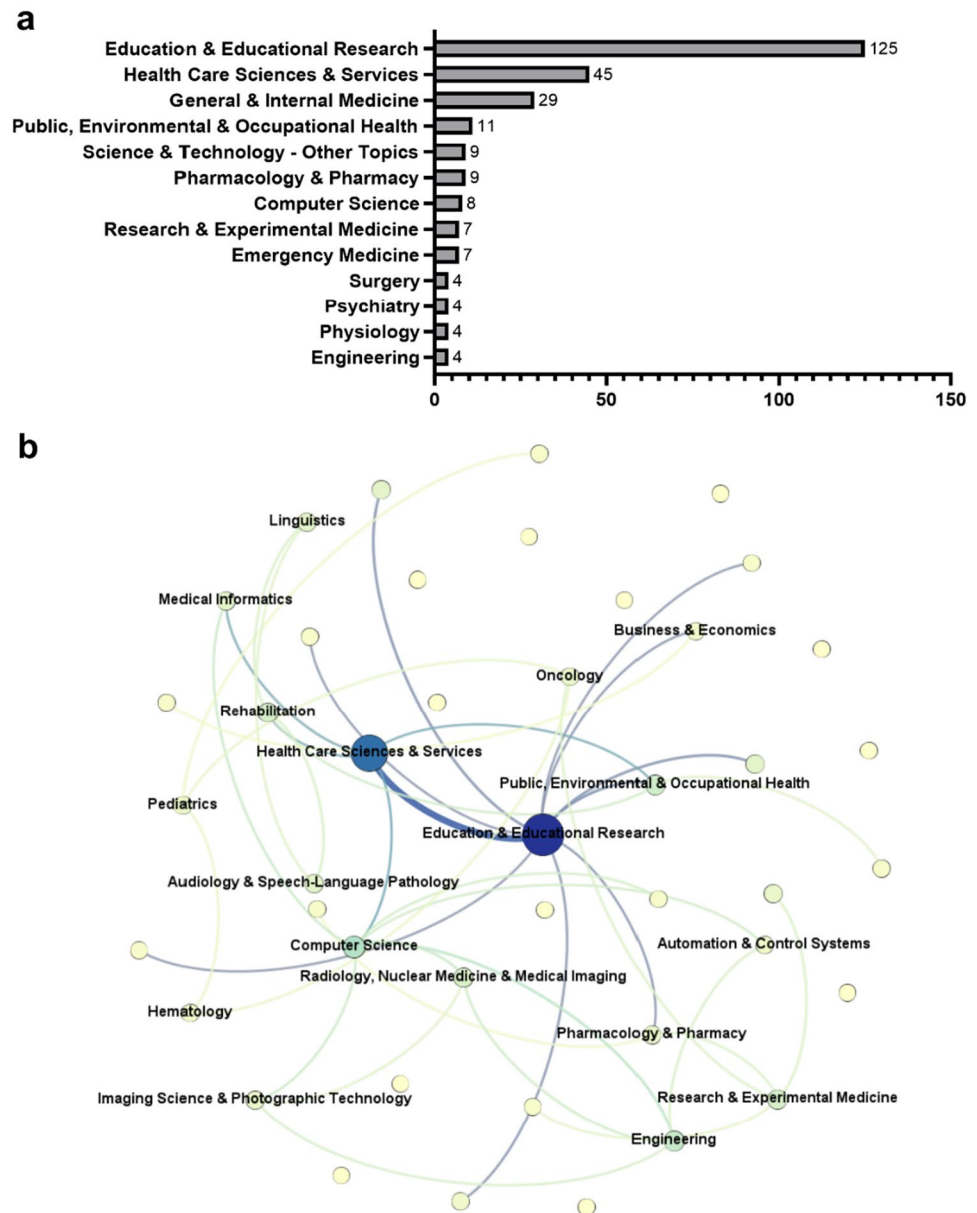


Fig. 2 Research areas (2013–2022). **a** Research areas with frequency greater than or equal to 4. **b** Undirected network of research areas consisting of 43 nodes with frequency greater than or equal to 4 and 39 edges. Only nodes with DC higher than the network average (1.814) were labeled. Node sizes and colors according to WDC—the larger the node and the darker its color (darkening towards blue), the higher its WDC. Edge thickness according to the number of co-occurrences between nodes. The Fruchterman-Reingold algorithm provides the layout of the network



to its DC (10), WDC (114.0), and EC (1.0). The research area Health Care Sciences & Services holds the second most central position, following the WDC (88.0) and EC (0.918). However, when considering DC, Computer Science is the second most central node (DC = 7). Although Computer Science is only the seventh most frequent research area (3.49%), it is connected to seven other nodes and ranks third in centrality according to EC (0.916). The highest co-occurrence between research areas is between Education & Educational Research and Health Care Sciences & Services (38 articles). General practice, pediatrics, psychiatry [30], and geriatrics [31] are examples of medical education fields included in some of these publications.

Figure 3a shows the most publishing countries on the subject. The United States of America (USA) ranks first,

with 33.62% of all publications, followed by China (8.73%) and Canada (7.42%). Together, they account for almost 50% of total publications. The country with the higher DC (11) and WDC (32) in the network of countries (Fig. 3b) is the Netherlands. Canada ranks second (DC = 10; WDC = 28), and the USA ranks third (DC = 8; WDC = 24.0). Although the USA and China lead in the number of publications, only 11.69% and 20% of their publications, respectively, involve collaboration with other countries. In comparison, 68.75% of the Netherlands' publications and 52.94% of Canada's publications are collaborations with other countries. Considering the EC, the Netherlands is the most central node (EC = 1.0), followed by Canada (EC = 0.917) and Australia (EC = 0.707). The collaborations between the Netherlands and Australia (three articles) and between Canada and the

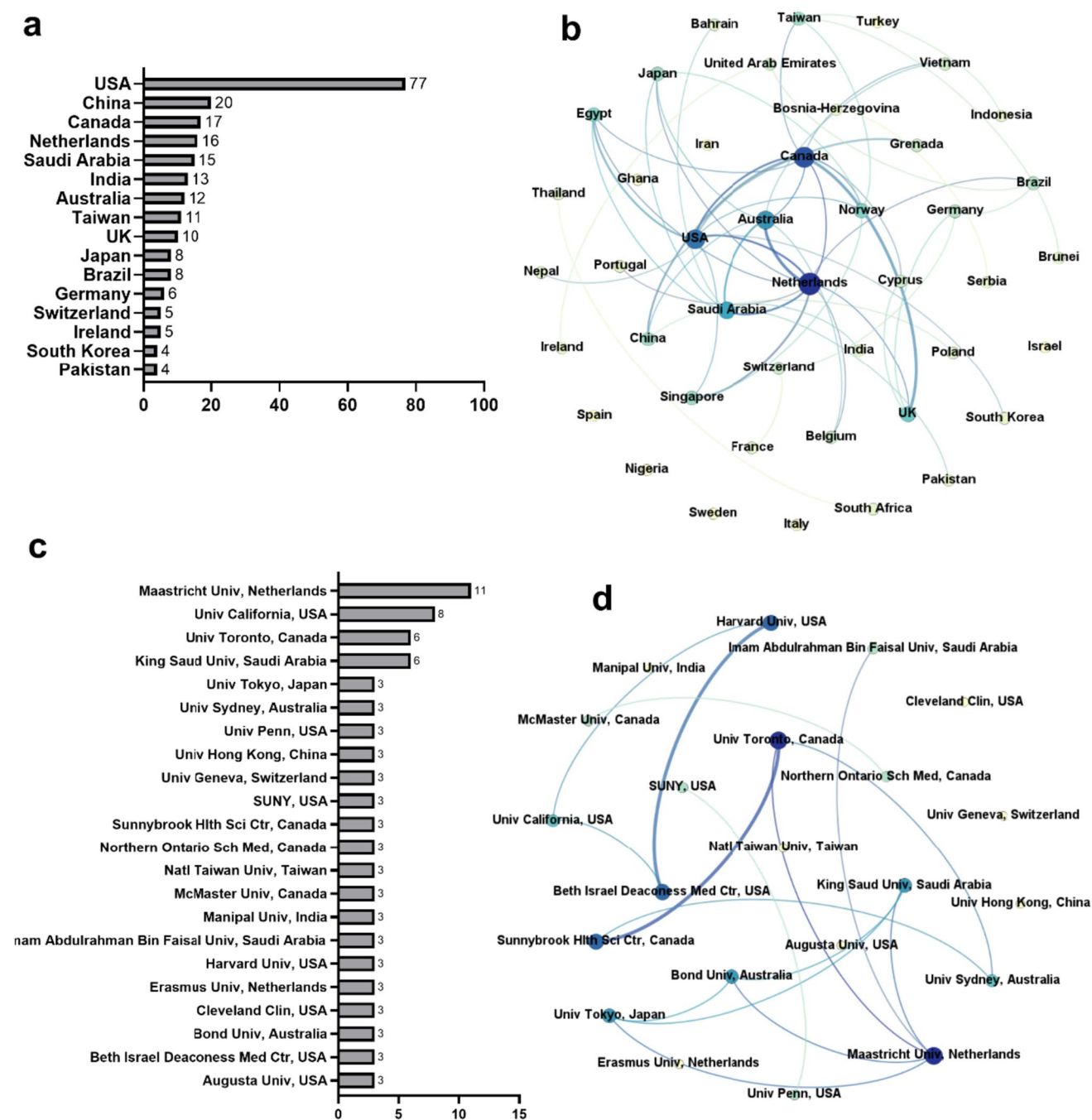


Fig. 3 Countries and organizations (all years—2013–2022). **a** Countries with four or more articles. **b** Undirected network of research areas consisting of 43 nodes with a frequency greater than or equal to 3 and 51 edges. **c** Universities with a frequency greater than or equal to 3. **d** Undirected network of organizations consisting of 22 nodes with a frequency equal to or greater than 3 and 16 edges. In both net-

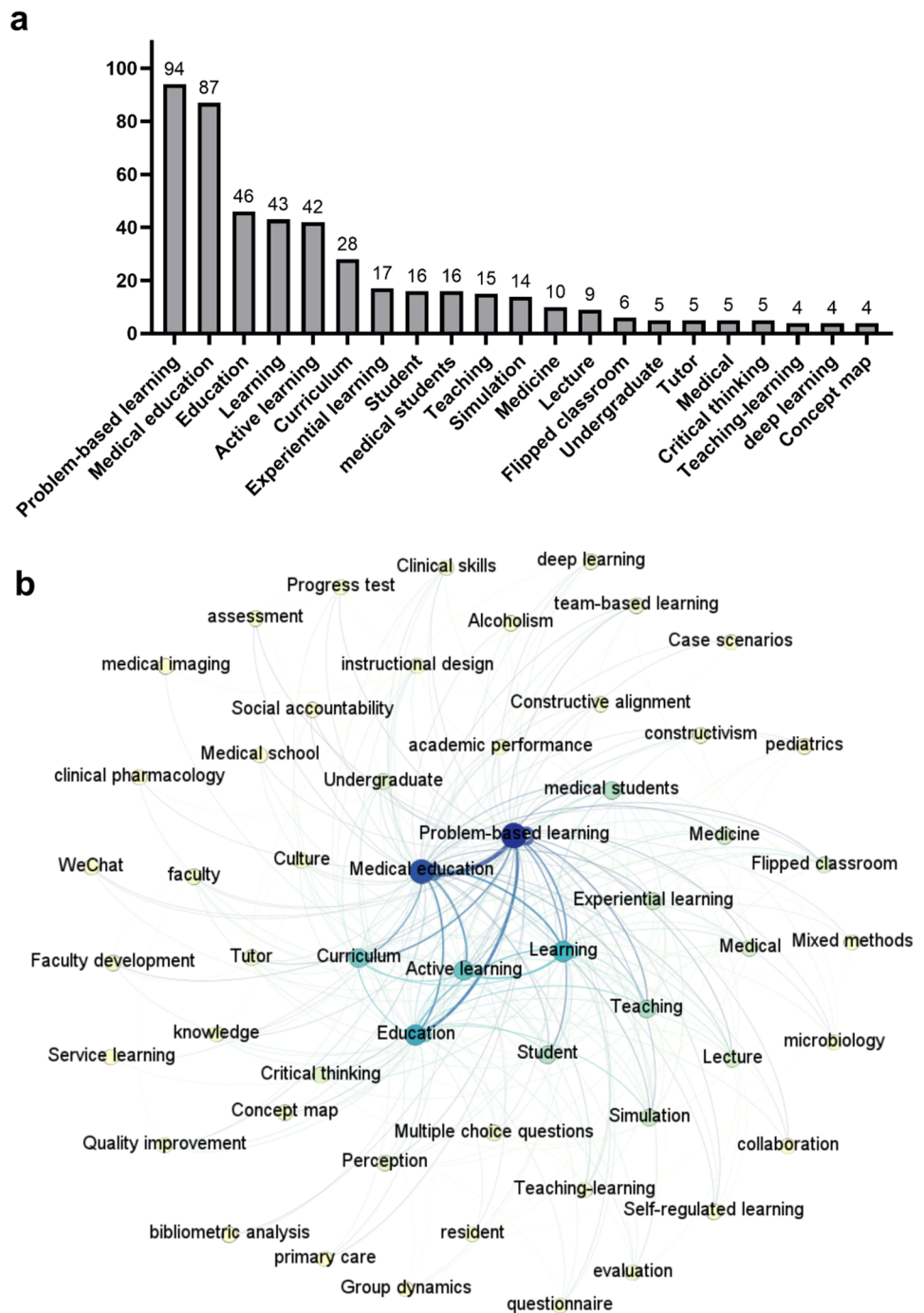
works (**b** and **d**), node sizes and colors are according to WDC—the larger the node and the darker its color (darkening towards blue), the higher its WDC. Edge thickness according to the number of co-occurrences between nodes. The Fruchterman-Reingold algorithm provides the layout of the two networks

UK (three articles) are the most prominent in the countries' network (Fig. 3b). The former includes articles on the assessment of PBL education [17], actions to maintain an effective PBL curriculum [32], and the effects of time on a medical PBL undergraduate curriculum [33]. The latter

primarily focuses on replacing paper-based patient cases with virtual patients in the PBL approach [34, 35].

Most organizations with three or more articles on PBL and medical education (Fig. 3c) are from the USA and Canada. However, the organization leading the rank is from the

Fig. 4 Keywords (all years—2013–2022). **a** Keywords with four or more articles. **b** Undirected network of keywords consisting of 43 nodes with a frequency greater than or equal to 4 and 55 edges. Node sizes and colors according to WDC—the larger the node and the darker its color (darkening towards blue), the higher its WDC. Edge thickness according to the number of co-occurrences between nodes. The Fruchterman-Reingold algorithm provides the layout of the network



Netherlands: Maastricht University (4.80%). Maastricht University is also the most central in the network of organizations (Fig. 3d) (DC = 5; WDC = 10; EC = 1.0). The University of Toronto in Canada holds the second most central position based on its DC (3) and WDC (10.0). However, according to the EC, the second position is shared by three organizations: the University of Tokyo (Japan; EC = 0.804), Bond University (Australia; EC = 0.804), and King Saudi University (Saudi Arabia; EC = 0.804). These three organizations have

collaborated with each other and with Maastricht University, which is the most central node. Their collaborative work resulted in a paper that provides guidelines for maintaining the effectiveness of PBL after its adoption [32]. Two other significant collaborations, resulting in three articles each, involve organizations within the same country. One collaboration is between Harvard University and the Beth Israel Deaconess Medical Center from the USA, and the other is between the University of Toronto and Sunnybrook Health Sciences Centre

from Canada. An example of a collaborative subject between American organizations is using concept maps to enhance PBL among first-year medical students [36]. For its part, a topic that the Canadian organizations collaborated on is experiential learning in quality improvement education [37, 38].

The most frequent authors' keywords are PBL (41.05%), medical education (37.99%), and education (20.09%) (Fig. 4a). In the network of keywords (Fig. 4b), PBL is the most central node (DC = 45; WDC = 440.0; EC = 1.0), followed by medical education (DC = 41; WDC = 390.0; EC = 0.930) and education (DC = 33; WDC = 246.0; EC = 0.858). Among the keywords not used in the search strategy, curriculum holds the most central position (DC = 28; WDC = 166.0; EC = 0.770), followed by teaching (DC = 18; WDC = 98.0; EC = 0.615) and students (DC = 17; WDC = 90.0; EC = 0.548). When examining co-occurrence patterns, PBL predominantly appears alongside medical education (45 articles), followed by education (30 articles) and curriculum (16 articles). Examples of PBL and medical education articles published in 2022 delve into subjects such as teaching neuroanatomy [39], internal medicine [40], and pathology [41]. Concerning articles published in 2022 with the keywords PBL and education, an example is the implementation of a digital PBL curriculum during the COVID-19 pandemic period [42]. For its part, an example of a 2022 article with the keywords PBL and curriculum is the introduction of a variation of PBL, named Patient-Oriented PBL [41].

Discussion

The findings of this study reveal several important aspects of the state of research in PBL in medical education from 2013 to 2022. The growth rate in the number of publications over the decade has been significant, despite a dip in 2020, possibly due to the global disruptions caused by the COVID-19 pandemic, which negatively affected research worldwide [43]. The recovery and upward trend in the subsequent years underscore the rising importance and recognition of PBL as a relevant educational strategy in medical training [13].

The top three journals publishing on PBL in medical education—*BMC Medical Education*, *Medical Teacher*, and *Medical Science Educator*—indicate the primary outlets for this research. Visiting the aims and scope sections of these journals, one can see that they focus on informing healthcare professionals about novel teaching strategies and methods, advancements in curriculum development, performance evaluations, assessments of training requirements, and the application of evidence-based medicine. Additionally, *Medical Teacher* is the official journal of the Association for Medical Education in Europe (AMEE), and *Medical Science Educator* is the official publication of the International Association of Medical Science Educators (IAMSE).

The prominence of Education & Educational Research among research areas comes as no surprise, given the pedagogical nature of PBL [11]. The strong link between this area and Health Care Sciences & Services underscores PBL's role at the intersection of education and healthcare provision [10, 19]. The high centrality of Computer Science, despite being the seventh most frequent research area, indicates an overlap of PBL research with digital technologies [44], suggestive of e-learning's growing importance in medical education [45]. Furthermore, the diverse application of PBL across various medical disciplines such as general practice [46], pediatrics [5], psychiatry [47], and geriatrics [48] suggests its adaptability, reinforcing its position as a versatile learning strategy in medical education.

Geographically, the USA leads in publications, followed by China and Canada, thus making North America and Asia the main contributors to PBL research in medical education. The significance of Canada in PBL was anticipated, given that McMaster University was the first to integrate PBL into its medical curriculum [1, 2]. In recent years, this university has been publishing on topics related to training residents in dementia care [49], critiquing the biomedical view as applied to PBL medical education [50], and examining the challenges faced by Asian countries in maintaining the PBL philosophy in health science education [51]. Equally predictable is the prominence of the USA, as most of its medical schools have embraced elements of PBL [1]. One of the most prestigious American universities is Harvard University. Authors affiliated with it have published on the use of concept maps to enhance PBL among first-year medical students [36], advancing experiential learning in systems-based practice and quality improvement initiatives [52], and medical active learning in rural Nepal [53]. As for China's participation, most Chinese medical schools use PBL for at least 50% of pre-clinical curricular hours, and PBL is seen as a tool for upgrading its healthcare system [54]. The University of Hong Kong is one of the prominent Chinese institutions where PBL is widely adopted. Authors from this institution have published on the perceptions of tutors and tutees from a near-peer teaching program [55], and on the assessment of clinical practice performance of speech-language pathology students trained with PBL tutorials [56, 57].

Although it ranks fourth in number of publications, the Netherlands shows a high degree of centrality in the network of countries, indicating it has established important collaboration with other countries. In the Netherlands, Maastricht University, in particular, plays a central role in the network of organizations. One of its prominent collaborators is Columbia University from the USA. Together, they have published a paper on global discourses and limitations of PBL in health education [58]. As known, Maastricht University was the second medical school in the world to

adopt PBL, and it is for quite some time now considered a role model in PBL education [59]. Beyond publications, Maastricht University has launched PBL training initiatives worldwide and held the Network of Community-Oriented Educational Institutions for Health Sciences secretariat. This global initiative, established by the World Health Organization (WHO), brings together educational institutions that practice PBL, further amplifying the university's influence and reach [19].

Regarding keywords, PBL and medical education are the most frequent and central in the network, which is expected considering the query we used in this investigation. Additionally, when examining keywords beyond our initial search criteria, the centrality of “curriculum,” “teaching,” and “students” demonstrates the importance of these elements in the application of PBL, suggesting that PBL is inherently connected to curriculum development, teaching strategies, and student outcomes [10, 60]. The co-occurrence patterns, coupled with the diversity of PBL applications, again indicate the versatility of PBL in responding to varied educational demands.

Conclusion

Our research findings underscore the growing significance and global interest in PBL in medical education. A 9.42% annual increase in PBL-related publications from 2013 to 2022 highlights its widespread acceptance, with leading journals like *BMC Medical Education*, *Medical Teacher*, and *Medical Science Educator* playing pivotal roles in disseminating advancements in teaching strategies, curriculum development, and evidence-based medicine. Considering research areas, there is a dominance of Education & Educational Research (54.58%), while strong connections with Health Care Sciences & Services and Computer Science reveal its intersectionality between education, healthcare, and digital technologies. The USA's leading role in publications (33.62%), followed by China and Canada, emphasizes North America and Asia as major contributors, but Maastricht University in the Netherlands exhibits a strong interconnectedness in the organization network, suggesting its prominent role in the field. Frequent keywords like PBL, medical education, and education form central nodes in the network, while additional keywords such as “curriculum,” “teaching,” and “students” underscore PBL's inherent connections to various educational elements. The diversity of contributors also hints at the rich cross-cultural and interdisciplinary exchanges possible within this field. These findings suggest that PBL may have a role to play in shaping future advancements in medical training, offering a potentially more integrative and pragmatic approach to learning.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s40670-024-02003-1>.

Author Contribution JG: conceptualization; formal analysis; investigation; methodology; writing—original draft. LB: conceptualization; formal analysis; investigation; methodology; writing—original draft. BC: conceptualization; methodology; writing—original draft; RM: conceptualization; methodology; writing—original draft; writing—review and editing. FM: conceptualization; data curation; formal analysis; investigation; methodology; software; supervision; writing—original draft; writing—review and editing.

Material and/or Code Availability The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics Approval Not applicable.

Consent to Participate Not applicable.

Consent to Publish Not applicable.

Competing Interests The authors declare no competing interests.

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